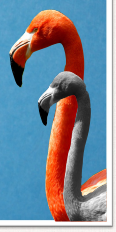
CHAPTER 2
Modeling Distributions of Data

2.2
Density Curves and Normal Distributions

The Practice of Statistics, 5th Edition
Starnes, Tabor, Yates, Moore



Bedford Freeman Worth Publishers

Density Curves and Normal Distributions

Learning Objectives

After this section, you should be able to:

- ✓ ESTIMATE the relative locations of the median and mean on a density curve.
- ✓ ESTIMATE areas (proportions of values) in a Normal distribution.
- ✓ FIND the proportion of z-values in a specified interval, or a z-score from a percentile in the standard Normal distribution.
- ✓ FIND the proportion of values in a specified interval, or the value that corresponds to a given percentile in any Normal distribution.
- ✓ DETERMINE whether a distribution of data is approximately Normal from graphical and numerical evidence.

Exploring Quantitative Data

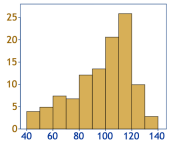
In Chapter 1, we developed a kit of graphical and numerical tools for describing distributions. Now, we'll add one more step to the strategy.

Exploring Quantitative Data

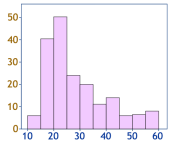
1. Always plot your data: make a graph, usually a dotplot, stemplot, or histogram.
2. Look for the overall pattern (shape, center, and spread) and for striking departures such as outliers.
3. Calculate a numerical summary to briefly describe center and spread.
4. Sometimes the overall pattern of a large number of observations is so regular that we can describe it by a smooth curve.

Data can be "distributed" (spread out) in different ways.

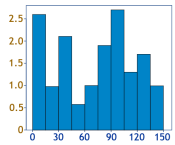
It can be spread out more on the left



Or more on the right



Or it can be all jumbled up



But there are many cases

Where the data tends to be around a central value with no bias left or right, and it gets close to a "Normal Distribution" like this:



The blue curve is a Normal Distribution. The yellow histogram shows some data that follows it closely, but not perfectly (which is usual).



It is often called a "Bell Curve" because it looks like a bell.

Many things closely follow a Normal Distribution:

- heights of people
- size of things produced by machines
- errors in measurements
- blood pressure
- marks on a test

Galton Board and the Normal Distribution

Quincunx

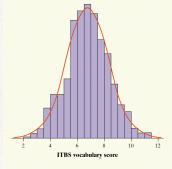
Density Curves 密度曲线

A density curve is a curve that
- is always on or above the horizontal axis, and
- has area exactly 1 underneath it.

A density curve describes the overall pattern of a distribution. The area under the curve and above any interval of values on the horizontal axis is the proportion of all observations that fall in that interval.

Example

The overall pattern of this histogram of the scores of all 947 seventh-grade students in Gary, Indiana, on the vocabulary part of the Iowa Test of Basic Skills (ITBS) can be described by a smooth curve drawn through the tops of the bars.



Describing Density Curves

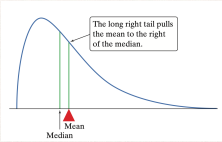
Our measures of center and spread apply to density curves as well as to actual sets of observations.

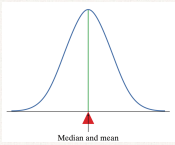
Distinguishing the Median and Mean of a Density Curve

The median of a density curve is the equal-areas point, the point that divides the area under the curve in half.

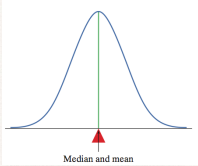
The mean of a density curve is the balance point, at which the curve would balance if made of solid material.

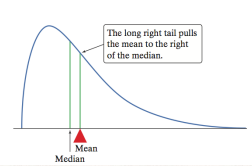
The median and the mean are the same for a symmetric density curve. They both lie at the center of the curve. The mean of a skewed curve is pulled away from the median in the direction of the long tail.





Describing Density Curves





- A density curve is an idealized description of a distribution of data.
- We distinguish between the mean and standard deviation of the density curve and the mean and standard deviation computed from the actual observations.
- The usual notation for the mean of a density curve is μ (the Greek letter mu). We write the standard deviation of a density curve as σ (the Greek letter sigma).

(I) Descriptive Statistics

$$\bar{x} = \frac{\sum x_i}{n}$$

$$s_x = \sqrt{\frac{1}{n-1}\sum\left(x_i - \bar{x}\right)^2}$$

The Standard Deviation

$$\text{variance} = s_x^2 = \frac{(x_1-\bar{x})^2+(x_2-\bar{x})^2+\ldots+(x_n-\bar{x})^2}{n-1} = \frac{1}{n-1}\sum(x_i-\bar{x})^2 \qquad \bar{x} = \frac{\Sigma xi}{n}$$

$$\text{standard deviation} = s_x = \sqrt{\frac{1}{n-1}\sum(x_i-\bar{x})^2}$$

(n-1: unbiased estimate 无偏估计)

$\bar{x}$ = the mean of a SAMPLE n = number of observations in a sample

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$$\text{population variance} = \sigma^2 = \frac{1}{N}\Sigma(x_i-\mu)^2 \qquad \mu = \frac{\Sigma xi}{N}$$

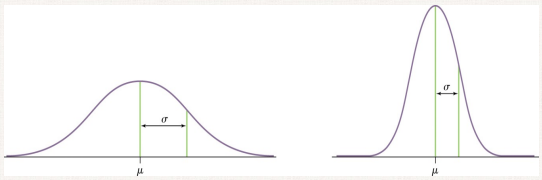
$$\text{population standard deviation} = \sigma = \sqrt{\frac{1}{N}\Sigma(x_i-\mu)^2}$$

μ = the mean of a POPUATION N = number of observations in a population

Normal Distributions 正态分布

One particularly important class of density curves are the Normal curves, which describe Normal distributions.

- All Normal curves have the same shape: symmetric 对称的, single-peaked 单峰的, and bell-shaped
- Any specific Normal curve is completely described by giving its mean μ（决定函数的位置）and its standard deviation σ（决定形状）.



Normal Distributions

A Normal distribution is described by a Normal density curve. Any particular Normal distribution is completely specified by two numbers: its mean μ and standard deviation σ.

- The mean of a Normal distribution is the center of the symmetric Normal curve.
- The standard deviation is the distance from the center to the change-of-curvature points on either side.
- We abbreviate the Normal distribution with mean μ and standard deviation σ as N(μ,σ).

Why are the Normal distributions important in statistics?

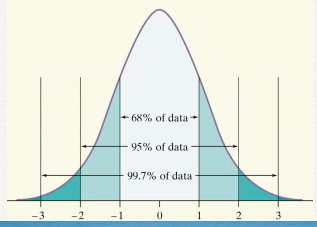
- Normal distributions are good descriptions for some distributions of real data.
- Normal distributions are good approximations of the results of many kinds of chance outcomes.
- Many statistical inference procedures are based on Normal distributions.

The 68-95-99.7 Rule/ Empirical Rule

Although there are many Normal curves, they all have properties in common.

The 68-95-99.7 Rule
In the Normal distribution with mean μ and standard deviation σ:

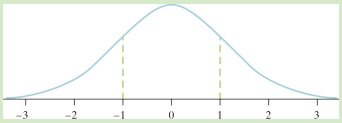
- Approximately 68% of the observations fall within σ of μ.
- Approximately 95% of the observations fall within 2σ of μ.
- Approximately 99.7% of the observations fall within 3σ of μ.



The Standard Normal Distribution 标准正态分布

All Normal distributions are the same if we measure in units of size σ from the mean μ as center.

The standard Normal distribution is the Normal distribution with mean 0 and standard deviation 1.



If a variable x has any Normal distribution N(μ,σ) with mean μ and standard deviation σ, then the standardized variable

$$z = \frac{x - \mu}{\sigma}$$

has the standard Normal distribution, N(0,1).

The Standard Normal Table

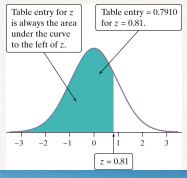
The standard Normal Table (Table A) is a table of areas under the standard Normal curve. The table entry for each value z is the area under the curve to the left of z.

Suppose we want to find the proportion of observations from the standard Normal distribution that are less than 0.81.

We can use Table A:

z	.00	.01	.02
0.7	.7580	.7611	.7642
0.8	.7881	.7910	.7939
0.9	.8159	.8186	.8212

P(z < 0.81) = .7910



Technology Corner

1. What proportion of observations from the standard Normal distribution are greater than −1.78?

   Recall that the standard Normal distribution has mean 0 and standard deviation 1.

2. What proportion of observations from the standard Normal distribution are between −1.25 and 0.81?

3. What is the 90th percentile of the standard Normal distribution?

Practice: Tiger on the Range

On the driving range, Tiger Woods practices his swing with a particular club by hitting many, many balls. Suppose that when Tiger hits his driver, the distance the ball travels follows a Normal distribution with mean 304 yards and standard deviation 8 yards.

Problem: What percent of Tiger's drives travel at least 290 yards?

Table A Standard normal probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367

Practice: Tiger on the Range

On the driving range, Tiger Woods practices his swing with a particular club by hitting many, many balls. Suppose that when Tiger hits his driver, the distance the ball travels follows a Normal distribution with mean 304 yards and standard deviation 8 yards.

Problem: What percent of Tiger's drives travel between 305 and 325 yards?

Table A (*Continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

AP® Statistics Practice Test

t2.1 Many professional schools require applicants to take a standardized test. Suppose that 1000 students take such a test. Several weeks after the test, Pete receives his score report: he got a 63, which placed him at the 73rd percentile. This means that

(a) Pete's score was below the median.
(b) Pete did worse than about 63% of the test takers.
(c) Pete did worse than about 73% of the test takers.
(d) Pete did better than about 63% of the test takers.
(e) Pete did better than about 73% of the test takers.

AP® Statistics Practice Test

t2.7 If the heights of a population of men follow a Normal distribution, and 99.7% have heights between 5′0″ and 7′0″, what is your estimate of the standard deviation of the heights in this population?

(a) 1″
(b) 3″
(c) 4″
(d) 6″
(e) 12″

AP® Statistics Practice Test

t2.8 Which of the following is not correct about a standard Normal distribution?

(a) The proportion of scores that satisfy $0 < z < 1.5$ is 0.4332.
(b) The proportion of scores that satisfy $z < -1.0$ is 0.1587.
(c) The proportion of scores that satisfy $z > 2.0$ is 0.0228.
(d) The proportion of scores that satisfy $z < 1.5$ is 0.9332.
(e) The proportion of scores that satisfy $z > -3.0$ is 0.9938.

Density Curves and Normal Distributions

Section Summary

In this section, we learned how to...

✓ESTIMATE the relative locations of the median and mean on a density curve.

✓ESTIMATE areas (proportions of values) in a Normal distribution.

✓FIND the proportion of z-values in a specified interval, or a z-score from a percentile in the standard Normal distribution.

✓FIND the proportion of values in a specified interval, or the value that corresponds to a given percentile in any Normal distribution.

✓DETERMINE whether a distribution of data is approximately Normal from graphical and numerical evidence.

Normal Distribution

The distribution of monthly rent for one-bedroom apartments in a city is approximately normal with mean $936 and standard deviation $61. A graduate student is looking for a one bedroom apartment and wants to pay no more than $800 in monthly rent. Of the following, which is the best estimate of the percent of one-bedroom apartments in the city with a monthly rent of at most $800 ?

The height of 3-year-old boys is approximately normally distributed. Duncan and Shane are 3-year-old boys. Duncan is 32.0 inches tall and is at the 32nd percentile of the distribution. Shane is 34.0 inches tall and is at the 62nd percentile of the distribution. Which of the following is closest to the mean of the height distribution?

The distribution of heights of 6-year-old girls is approximately normally distributed with a mean of 46.0 inches and a standard deviation of 2.7 inches. Aliyaah is 6 years old, and her height is 0.96 standard deviation above the mean. Her friend Jayne is also 6 years old and is at the 93rd percentile of the height distribution. At what percentile is Aliyaah's height, and how does her height compare to Jayne's height?

(A) Aliyaah's height is at the 17th percentile of the distribution, and she is shorter than Jayne.
(B) Aliyaah's height is at the 67th percentile of the distribution, and she is shorter than Jayne.
(C) Aliyaah's height is at the 67th percentile of the distribution, and she is taller than Jayne.
(D) Aliyaah's height is at the 83rd percentile of the distribution, and she is shorter than Jayne.
(E) Aliyaah's height is at the 83rd percentile of the distribution, and she is taller than Jayne.

The table shows whether students in an introductory statistics class like dogs and/or cats.

		Like Dogs		
		Yes	No	Total
Like Cats	Yes	194	21	215
	No	110	10	120
	Total	304	31	335

a. What is the marginal distribution (in %) of "liking dogs"?

b. What is the conditional distribution (in %) of "liking dogs" for students who like cats?

c. What kind of display(s) would you use to show the association between "liking dogs" and "liking cats"? (Just name the graph(s).)

